

WHEN GAUSSIAN NOISE STOPS BEING SMALL: SCHATTEN FILTERING FOR RESOLUTION-STABLE LEARNING

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ABSTRACT

Gaussian noise injection is ubiquitous in representation learning, neural operators, and diffusion models. Yet a noise level that is benign at one resolution need not represent the same perturbation after refinement: its relative covariance update can accumulate spectral mass in newly resolved modes, causing Gaussian objectives and their gradients to be governed by the discretization rather than signal geometry. We formulate this failure as a continuum-stability problem. For a whitened covariance perturbation K_N , we show that uniform Hilbert–Schmidt control is the sharp threshold for uniform boundedness of Gaussian noise objectives under refinement and fixed spectral bounds. Trace-class perturbations admit ordinary trace and log-determinant formulas; Hilbert–Schmidt but non-trace-class perturbations require the Carleman–Fredholm determinant $\det_2$; beyond $\mathcal{S}_2$, this second-order-renormalized objective loses uniform cutoff control. This characterization yields *Schatten-filtered noise*, a spectral gate that explicitly constrains the perturbation to a chosen operator-ideal regime. Synthetic power-law spectra recover the predicted three-phase boundary, while resolution-scaling tests measure objective drift, gradient variance, and noise-scale transfer. Finally, we ask whether latent encoders act as implicit operator-ideal filters by auditing both pushed-forward input perturbations and downstream diffusion updates across resolutions. The resulting framework turns a hidden requirement of high-dimensional noise injection into a falsifiable representation diagnostic and a controllable noise-design principle.

1 INTRODUCTION

Gaussian noise is routinely used as a scalar control knob: choose a variance, inject a perturbation, and tune its strength against predictive or generative performance. This picture underlies feature-space regularization, stochastic latent representations, neural operators, and diffusion models. It is reliable at a fixed dimension, where every covariance matrix and Gaussian objective is well defined. It becomes misleading when dimension represents resolution. Refinement introduces additional modes, and a perturbation with unchanged nominal strength can accumulate spectral mass in every newly resolved direction. The same hyperparameter then ceases to describe the same geometric intervention. A penalty intended to measure meaningful signal–noise geometry can instead be dominated by the discretization cutoff, while its gradients can transmit the same resolution dependence through the parameter-sensitivity spectrum.

We make this failure precise through the relative covariance perturbation. Let C_N be a reference covariance at resolution N , let $C_N + \Delta C_N$ be its perturbed counterpart, and define

$$K_N = C_N^{-1/2} \Delta C_N C_N^{-1/2}. \quad (1)$$

For $I + K_N \succ 0$, the Gaussian covariance residue has the spectral form

$$\mathcal{L}_N = \frac{1}{2} \sum_{j \leq n_N} [\lambda_j(K_N) - \log(1 + \lambda_j(K_N))]. \quad (2)$$

Because $\lambda - \log(1 + \lambda) = \lambda^2/2 + O(\lambda^3)$ near zero, the unresolved tail is governed by squared spectral mass. This exposes the finite-dimensional blind spot: every finite matrix belongs to every

Schatten class, so membership at one N says nothing about refinement. The relevant quantity is the dimension-uniform diagnostic

$$T_2(N) := \|K_N\|_{\mathcal{S}_2}^2, \quad \sup_N T_2(N) < \infty. \quad (3)$$

This criterion leads directly to *Schatten-filtered noise*. We apply a spectral gate $K_N \mapsto K_N^{(\tau)} = \Phi_\tau(K_N)$ whose attenuation is chosen to place the relative perturbation in a prescribed operator-ideal regime uniformly over resolution. In the trace-class regime $\mathcal{S}_1$, ordinary trace and log-determinant expressions remain available. In the larger regime $\mathcal{S}_2 \setminus \mathcal{S}_1$, those terms need not exist separately, but their finite combination is captured by the Carleman–Fredholm determinant,

$$\mathcal{L}_N^{\text{ren}} = -\frac{1}{2} \log \det_2(I + K_N). \quad (4)$$

Outside $\mathcal{S}_2$, the renormalized tail itself is not cutoff stable, so filtering must alter the perturbation spectrum rather than merely rewrite the objective. The method is therefore ideal-aware rather than simply frequency-aware: it controls accumulated relative covariance geometry after whitening, including anisotropy and spectral multiplicity.

Our main result identifies $\mathcal{S}_2$ as the sharp stability boundary. Under dimension-independent positivity and spectral bounds, $T_2(N)$ is equivalent up to fixed constants to the Gaussian covariance residue, and it exactly determines the variance of the centered sample log-density ratio. Consequently, bounded $T_2(N)$ yields an objective uniformly controlled under refinement, whereas diverging $T_2(N)$ forces unbounded cutoff growth. For spectra $s_j(K) \asymp j^{-\alpha}$, this gives a three-phase law: bounded behavior for $\alpha > 1/2$, logarithmic growth at $\alpha = 1/2$, and polynomial growth for $\alpha < 1/2$.

Diffusion supplies the representation-level question. Latent diffusion should not be viewed merely as computational compression: an encoder maps input perturbations and defines the reference covariance relative to which downstream Gaussian noise acts. Rather than presuming that compression supplies sufficient regularity, we ask whether learned encoders control both relative covariance families under refinement. Whitened Jacobian spectra, raw $\mathcal{S}_2$ mass, and pixel-space, isometric, and learned-encoder controls make this hypothesis falsifiable. Schatten filtering enforces the hidden requirement explicitly instead of relying on representation compression to provide it implicitly.

Our contributions are:

- **A sharp continuum-stability criterion.** We formulate Gaussian noise injection as a resolution-indexed relative covariance problem and identify Schatten-2 control as the cutoff-stability threshold, including the bounded, logarithmically critical, and polynomial power-law regimes.
- **An operator-ideal-aware noise mechanism.** We introduce Schatten-filtered noise together with the appropriate $\det_2$ objective in the Hilbert–Schmidt but non-trace-class regime. Its criterion targets cumulative covariance geometry rather than coordinate frequency alone.
- **Multi-scale tests connecting theory to learning.** Controlled spectra test the sharp boundary, operator-learning tasks test noise-scale transfer under grid refinement, and a Jacobian pushforward diagnostic asks whether latent encoders control tail $\mathcal{S}_2$ mass.

2 GAUSSIAN NOISE UNDER REFINEMENT

2.1 RESOLUTION-INDEXED PERTURBATIONS

Let n_N denote the number of active coordinates or modes at resolution N , with $n_N \rightarrow \infty$. We compare the centered Gaussian reference and its noise-perturbed counterpart

$$p_N = \mathcal{N}(0, C_N), \quad q_N = \mathcal{N}(0, C_N + \Delta C_N). \quad (5)$$

We assume $C_N \succ 0$ and $C_N + \Delta C_N \succ 0$ on the active subspace. Whitening by C_N gives the self-adjoint relative covariance perturbation K_N from Equation (1), so that q_N has relative covariance $I + K_N$. The eigenvalues of K_N measure noise strength relative to the reference geometry rather than in raw coordinates.

For finite n_N , the covariance contribution to the Gaussian KL divergence is

$$D_{\text{KL}}(q_N \| p_N) = \frac{1}{2} (\text{Tr } K_N - \log \det(I + K_N)) = \frac{1}{2} \sum_{j \leq n_N} g(\lambda_{j,N}), \quad (6)$$

where $g(\lambda) = \lambda - \log(1 + \lambda)$. Every term in Equation (6) is finite at every fixed resolution. Stability is therefore a property of the sequence $(K_N)_N$, not of any individual matrix.

Some implementations specify the same update through a relative precision perturbation H_N ,

$$q_N = \mathcal{N}\left(0, C_N^{1/2}(I - H_N)^{-1}C_N^{1/2}\right), \quad K_N = H_N(I - H_N)^{-1}. \quad (7)$$

Conversely, $H_N = K_N(I + K_N)^{-1}$. Under a dimension-independent positivity gap and operator-norm bound, these spectral maps preserve uniform $\mathcal{S}_2$ membership up to fixed constants. We state the theory in the additive-covariance parameterization K_N , while using Equation (7) to evaluate precision-parameterized noise layers.

2.2 HOW NOMINALLY SMALL NOISE BECOMES LARGE

A scalar covariance rescaling already exhibits the issue. If $C_N + \Delta C_N = aC_N$ for a fixed $a \neq 1$, then

$$K_N = (a - 1)I_{n_N}, \quad T_2(N) = (a - 1)^2 n_N. \quad (8)$$

The perturbation is harmless at each cutoff but grows in every newly resolved direction. In the infinite-rank limit, the identity-direction update is not Hilbert–Schmidt; this is the operator-theoretic reason that scalar covariance changes can leave a Gaussian equivalence class (Bogachev, 1998). For learning, the immediate consequence is simpler: a nominal noise scale can become a resolution-dependent intervention.

The same accumulation appears for decaying spectra. If $s_j(K) \asymp j^{-\alpha}$, then the pointwise perturbation vanishes at high frequency for every $\alpha > 0$, but its cumulative squared mass may still diverge. Thus “low amplitude in every tail mode” and “stable under refinement” are different statements.

2.3 THE CORRECT STABILITY OBJECT

We call a family of Gaussian perturbations *uniformly refinement stable* when its renormalized covariance objective and the L^2 norm of its centered sample log-density ratio remain bounded uniformly over N . This notion is uniform control, not by itself convergence of an arbitrary sequence of cutoffs. The diagnostic is

$$T_2(N) = \text{Tr}(K_N^* K_N) = \sum_{j \leq n_N} s_j(K_N)^2. \quad (9)$$

This definition does not ask whether a finite matrix is Schatten: it asks whether the Schatten mass stays bounded along the entire refinement path.

When $K_N \in \mathcal{S}_1$ uniformly, the trace and log determinant in Equation (6) are separately uniformly controlled. For compatible Galerkin cutoffs of a fixed trace-class operator, they also converge separately. The larger $\mathcal{S}_2$ regime requires their combined finite part,

$$-\log \det_2(I + K_N) = \text{Tr } K_N - \log \det(I + K_N), \quad (10)$$

which is an exact identity at every finite cutoff. The distinction becomes visible only as N grows: $\mathcal{S}_1$ controls the two ordinary components, whereas $\mathcal{S}_2$ controls their renormalized combination. In infinite dimensions, $\det_2(I + K)$ is defined directly for $K \in \mathcal{S}_2$ with $I + K \succ 0$; Equation (10) must not be read as subtracting two separately infinite quantities.

3 SCHATTEN-FILTERED NOISE

3.1 A NOISE-LAYER INTERFACE AND IDEAL-AWARE GATE

At a Gaussian noise-injection site, the raw relative covariance $K_N \succeq 0$ defines $\Delta C_N = C_N^{1/2} K_N C_N^{1/2}$. Schatten-filtered noise replaces it with $K_N^{(\tau)} = \Phi_\tau(K_N)$ and injects

$$\xi_N^{(\tau)} \sim \mathcal{N}\left(0, C_N^{1/2} K_N^{(\tau)} C_N^{1/2}\right), \quad z' = z + \xi_N^{(\tau)}. \quad (11)$$

A basis-invariant budget gate is the radial projection

$$\Phi_\tau^{\text{budget}}(K_N) = \min \left\{ 1, \frac{\tau}{\|K_N\|_{\mathcal{S}_2}} \right\} K_N, \quad (12)$$

with the multiplier defined as one when $K_N = 0$. It enforces $\|K_N^{(\tau)}\|_{\mathcal{S}_2} \leq \tau$ without choosing a coordinate basis.

Structured operators can preserve leading physical modes more selectively. Let $K_N = U_N \text{diag}(\lambda_{1,N}, \dots, \lambda_{n_N,N}) U_N^*$ in a refinement-compatible mode basis. We use

$$K_N^{(\tau)} = \Phi_\tau(K_N) := U_N \text{diag}(\phi_{\tau,1}(\lambda_{1,N}), \dots, \phi_{\tau,n_N}(\lambda_{n_N,N})) U_N^*. \quad (13)$$

with a nonnegative, resolution-independent envelope $w = (w_j)_{j \geq 1}$,

$$\phi_{\tau,j}(\lambda) = \text{sign}(\lambda) \min\{|\lambda|, \tau w_j\}. \quad (14)$$

where physical mode labels are fixed across refinements and the same cap is shared within symmetry or degeneracy blocks. If $w \in \ell^2$, then $\sup_N \|K_N^{(\tau)}\|_{\mathcal{S}_2} \leq \tau \|w\|_2$; if $w \in \ell^1$, the same construction targets trace class. Unlike a cutoff at a fixed fraction of available Fourier modes, the envelope is tied to a summability budget that does not expand with the grid.

In the Hilbert–Schmidt regime, we evaluate the covariance residue through

$$\mathcal{L}_N^{\text{ren}}(K_N) := -\frac{1}{2} \log \det_2(I + K_N). \quad (15)$$

For trace-class K_N , this equals the ordinary expression in Equation (6); for $K_N \in \mathcal{S}_2 \setminus \mathcal{S}_1$, it remains meaningful even when the trace and log determinant do not exist separately. The surrounding model can optimize the task loss on z' together with $\lambda \mathcal{L}_N^{\text{ren}}(K_N^{(\tau)})$ as an explicit resolution-stable noise budget. The theorem concerns this Gaussian covariance term; the task loss need not itself be Gaussian.

3.2 THE SHARP HILBERT–SCHMIDT THRESHOLD

Theorem 3.1 (Refinement stability of Gaussian noise). *Suppose $K_N = K_N^* \succeq 0$ and $\sup_N \|K_N\|_{\text{op}} \leq M < \infty$. Then*

$$\frac{1}{4(1+M)} T_2(N) \leq \mathcal{L}_N^{\text{ren}}(K_N) \leq \frac{1}{4} T_2(N). \quad (16)$$

Moreover, if $\ell_N(X) = \log \frac{dq_N}{dp_N}(X)$ with $X \sim q_N$, then

$$\text{Var}_{q_N}[\ell_N(X)] = \frac{1}{2} T_2(N). \quad (17)$$

Consequently, the following are equivalent:

$$\sup_N T_2(N) < \infty, \quad \sup_N \mathcal{L}_N^{\text{ren}}(K_N) < \infty, \quad \sup_N \text{Var}_{q_N}[\ell_N(X)] < \infty. \quad (18)$$

If $T_2(N) \rightarrow \infty$, both the renormalized covariance objective and the sample log-density variance diverge. The same equivalence holds for signed self-adjoint perturbations under a uniform positivity gap $I + K_N \succeq \delta I$ and a uniform operator-norm bound, with constants that depend only on (δ, M) .

The result makes the slogan precise: Schatten membership at a cutoff is automatic, but dimension-uniform Schatten control predicts whether Gaussian noise objectives survive refinement.

Uniform control also yields genuine cutoff consistency for compatible approximations. If $K_N \rightarrow K$ in $\mathcal{S}_2$ on a common Hilbert space and $I + K_N, I + K \succeq \delta I$, continuity of the regularized determinant gives

$$\mathcal{L}_N^{\text{ren}}(K_N) \longrightarrow -\frac{1}{2} \log \det_2(I + K). \quad (19)$$

Under a common isonormal Gaussian coupling, the centered log-density ratios converge in L^2 , with squared distance $\frac{1}{2} \|K_N - K\|_{\mathcal{S}_2}^2$. Thus uniform $\mathcal{S}_2$ boundedness supplies the sharp no-divergence criterion, while $\mathcal{S}_2$ convergence supplies a continuum limit.

Corollary 3.2 (Power-law phase transition). *Under the conditions of Theorem 3.1, if $s_j(K) \asymp j^{-\alpha}$ and K_N retains the first n_N modes, then*

$$T_2(N) \asymp \begin{cases} 1, & \alpha > \frac{1}{2}, \\ \log n_N, & \alpha = \frac{1}{2}, \\ n_N^{1-2\alpha}, & \alpha < \frac{1}{2}. \end{cases} \quad (20)$$

The renormalized objective and sample log-density variance inherit the same bounded, logarithmic, and polynomial regimes.

3.3 MATRIX-FREE DIAGNOSIS AND RENORMALIZATION

For independent unnormalized Rademacher probes $v_1, \dots, v_m \in \{-1, +1\}^{n_N}$,

$$\widehat{T}_2(N) = \frac{1}{m} \sum_{r=1}^m \|K_N v_r\|_2^2, \quad \mathbb{E} \widehat{T}_2(N) = T_2(N). \quad (21)$$

Thus the stability diagnostic needs only operator–vector products, not an eigendecomposition. When $\|K_N\|_{\text{op}} < 1$, the determinant term also admits the matrix-free expansion

$$-\log \det_2(I + K_N) = \sum_{k=2}^{\infty} \frac{(-1)^k}{k} \text{Tr}(K_N^k), \quad (22)$$

whose traces can reuse the same probes (Hutchinson, 1989). Our series-based implementation caps $\|K_N\|_{\text{op}} < 1$; a Fourier-diagonal gate can instead evaluate the regularized determinant exactly mode by mode.

What T_2 does and does not predict. Theorem 3.1 gives an exact statement for sample log-density fluctuations. For a parameterized family $K_N(\theta)$, gradient fluctuations also depend on the sensitivity spectrum $\partial_\theta K_N$; they are not a function of $T_2(N)$ alone. Our optimization experiments therefore report the source of randomness explicitly. The matrix-free test varies estimator probes, while the learning tests separately measure minibatch-gradient variance and noise-scale drift.

3.4 ENCODERS AS A FALSIFIABLE FILTERING HYPOTHESIS

Let $J_N(x)$ be an encoder Jacobian, let Σ_N be an input perturbation covariance, and let $C_{z,N}^{\text{ref}} \succ 0$ define an independently specified latent reference geometry. The whitened Jacobian and relative latent covariance update are

$$F_N(x) = (C_{z,N}^{\text{ref}})^{-1/2} J_N(x) \Sigma_N^{1/2}, \quad K_{E,N}(x) = F_N(x) F_N(x)^*. \quad (23)$$

Their Schatten masses obey the exact identity

$$T_2^{(E)}(N; x) := \|K_{E,N}(x)\|_{\mathcal{S}_2}^2 = \sum_j s_j(F_N(x))^4 = \|F_N(x)\|_{\mathcal{S}_4}^4. \quad (24)$$

Thus an encoder filters an input perturbation into a uniformly Hilbert–Schmidt covariance family exactly when its whitened Jacobian is uniformly Schatten-4. We test either $\sup_N T_2^{(E)}(N; x) < \infty$ pointwise or $\sup_N \mathbb{E}_x T_2^{(E)}(N; x) < \infty$ at the population level, without changing the reference covariance to the perturbation being audited. Compression alone does not establish this property: a rank- r_N partial isometry has $T_2^{(E)} = r_N$, which diverges whenever the effective rank grows.

The pushforward in Equation (23) is distinct from the relative covariance update used by a downstream diffusion step,

$$K_{\text{diff},N,t} = (C_{z,N}^{\text{ref}})^{-1/2} (C_{z,N,t} - C_{z,N}^{\text{ref}}) (C_{z,N}^{\text{ref}})^{-1/2}. \quad (25)$$

For a variance-preserving update $C_{z,N,t} = \alpha_t^2 C_{z,N}^{\text{ref}} + \sigma_t^2 I$, this object becomes

$$K_{\text{diff},N,t} = (\alpha_t^2 - 1) I + \sigma_t^2 (C_{z,N}^{\text{ref}})^{-1}, \quad (26)$$

which directly exposes how the latent reference spectrum changes the class in which an otherwise isotropic noise schedule acts. We audit both objects rather than assuming that attenuation of input perturbations automatically regularizes noise injected after encoding. This makes the latent-diffusion interpretation falsifiable and separates spectral filtering from a fixed-dimensional bottleneck. All adjoints and singular values use resolution-consistent Hilbert inner products, including discretization quadrature weights.

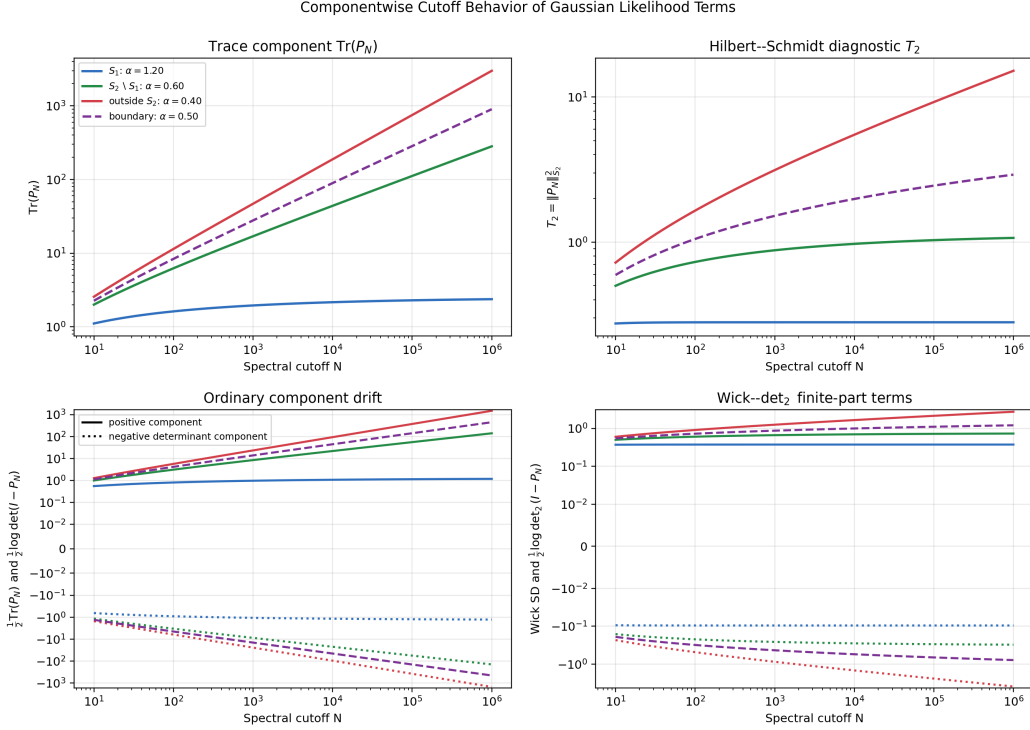


Figure 1: **The operator-ideal boundary appears only under refinement.** Power-law precision spectra traverse $\mathcal{S}_1$, $\mathcal{S}_2 \setminus \mathcal{S}_1$, the critical boundary, and the non- $\mathcal{S}_2$ regime. Ordinary trace and determinant components drift separately, while the Wick- $\det_2$ finite part and its stochastic fluctuation are controlled exactly when $\|H_N\|_{\mathcal{S}_2}^2$ is controlled.

4 EXPERIMENTS AS CLAIM TESTS

Each experiment has one inferential role: verify the sharp boundary, show why ideal membership is not a frequency heuristic, connect the diagnostic to a learning-relevant instability, or test the representation hypothesis. The primary axis is refinement at matched task and noise budgets, not isolated single-resolution accuracy.

4.1 THE THRESHOLD PREDICTS OBJECTIVE AND ESTIMATOR FLUCTUATIONS

We first use relative precision perturbations $\lambda_j(H_N) = 0.45j^{-\alpha}$ and increase the cutoff from 10 to 10^6 . The setting $\alpha = 1.20$ exposes ordinary trace-class behavior; $\alpha = 0.60, 0.50, 0.40$ isolate the bounded, logarithmic, and polynomial $\mathcal{S}_2$ regimes in Theorem 3.2. The covariance map $K_N = H_N(I - H_N)^{-1}$ from Equation (7) preserves these regimes. At every finite cutoff, with $Z \sim \mathcal{N}(0, I)$, we verify

$$\frac{1}{2} \langle Z, H_N Z \rangle + \frac{1}{2} \log \det(I - H_N) = \frac{1}{2} (\langle Z, H_N Z \rangle - \text{Tr } H_N) + \frac{1}{2} \log \det_2(I - H_N). \quad (27)$$

The two sides agree to numerical precision, while only the combined Wick- $\det_2$ expression remains uniformly controlled in the Hilbert-Schmidt, non-trace-class regime. Figure 1 shows the complete refinement path rather than a single-cutoff snapshot.

The same panel tests the stochastic consequence using the matrix-free objective. We hold the feature batch fixed and vary only its Rademacher probes. Across a $64\times$ increase in dimension, the unfiltered branch increases $\|H_N\|_{\mathcal{S}_2}^2$ by $42.5\times$ and probe-induced gradient variance by $102.7\times$; the Schatten-filtered branch changes by $1.21\times$ and $1.46\times$, respectively. This is an estimator-mechanism test, not a claim that T_2 alone determines every source of SGD variance.

4.2 SAME FREQUENCY APPEARANCE, DIFFERENT CONTINUUM FATE

The decisive control constructs two perturbations that a spatial-frequency heuristic cannot distinguish. Let $d = m^2$, let $0 < \eta < 1$, and let U_d be a flat DFT unitary in complex Fourier coordinates, so $|(U_d)_{ij}|^2 = 1/d$ (a Walsh–Hadamard transform gives a real construction at compatible d). Define

$$A_d^{\text{stable}} = U_d \text{diag} \left(\eta, \underbrace{\frac{\eta}{m+1}, \dots, \frac{\eta}{m+1}}_{d-1} \right) U_d^*, \quad (28)$$

$$A_d^{\text{unstable}} = U_d \text{diag} \left(\underbrace{\eta, \dots, \eta}_m, \underbrace{0, \dots, 0}_{d-m} \right) U_d^*. \quad (29)$$

They have exactly the same coordinatewise noise power, total noise power, and leading-mode gain:

$$\text{diag } A_d^{\text{stable}} = \text{diag } A_d^{\text{unstable}} = \frac{\eta}{m} \mathbf{1}, \quad \text{Tr } A_d^{(\cdot)} = \eta m, \quad \|A_d^{(\cdot)}\|_{\text{op}} = \eta. \quad (30)$$

Their operator-ideal behavior is nevertheless opposite:

$$\|A_d^{\text{stable}}\|_{S_2}^2 = \eta^2 \frac{2m}{m+1} \rightarrow 2\eta^2, \quad \|A_d^{\text{unstable}}\|_{S_2}^2 = \eta^2 m = \eta^2 \sqrt{d} \rightarrow \infty. \quad (31)$$

We attach the same physical Fourier envelope to both families. Their radial power spectrum, bandwidth, trace, operator norm, and low-frequency response then remain matched, but their raw T_2 and Gaussian objective fluctuations separate with the rates in Equation (31). A frequency-only mask acts on the common envelope and cannot remove the unstable channel factor without also sending the retained low-frequency response to zero. The main comparison includes hard and smooth Fourier cutoffs, an unconstrained spectral decay gate, and the ideal-aware gate at matched task distortion and noise budget. Exact construction and matching constraints are in Section C.

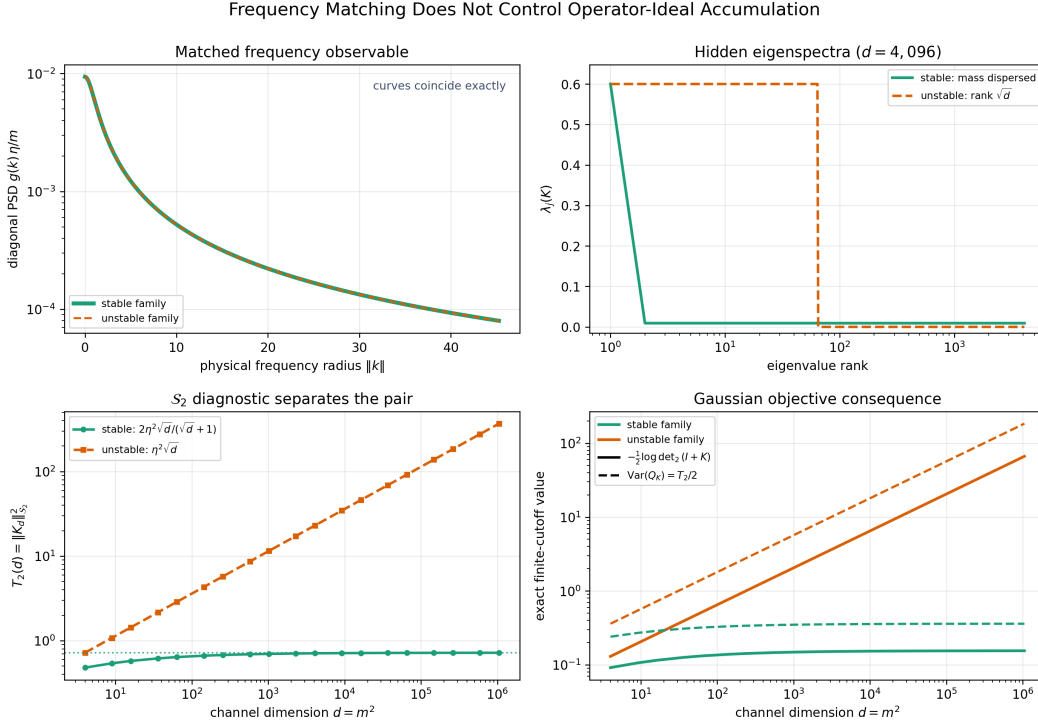


Figure 2: **Frequency matching does not control operator-ideal accumulation.** The two families have identical radial Fourier diagonals, trace, and operator norm, but different hidden eigenvalue multiplicities. At $d = 1,048,576$, the stable family has $T_2 = 0.7193$, while the unstable family reaches 368.64, a $512.5\times$ separation inherited by the exact Gaussian residue and centered quadratic variance. All curves are closed-form; explicit DFT matrices verify the identities at small d .

4.3 RESOLUTION-CONSISTENT OPERATOR LEARNING

We discretize the same underlying continuous random fields on a resolution ladder and train a neural operator at a base grid. The ideal-aware gate is defined in physical Fourier modes,

$$\beta_\theta(k) = \rho \operatorname{sigmoid}(a_\theta(k))(1 + \|k\|^2)^{-s/2}, \quad s = 1 + \epsilon + \operatorname{softplus}(\gamma), \quad (32)$$

where $0 < \rho < 1$ and $\epsilon > 0$ are fixed. Thus $s > 1$ enforces square summability in two spatial dimensions and keeps the series estimator inside its spectral-radius domain. Here $\beta_\theta(k)$ is a relative covariance eigenvalue, not a noise standard deviation. We freeze the selected gate and evaluate zero-shot at $2\times$ and $4\times$ the training resolution. Architecture, optimization steps, low-frequency response, and base-resolution noise budget are held fixed.

Baselines comprise ordinary Gaussian noise, renormalization without filtering, relative and fixed physical Fourier cutoffs, and an architecture-matched smooth decay gate without the $s > 1$ constraint. The primary outputs are raw canonical $T_2(R)$, minibatch-gradient variance, the drift of the independently selected noise scale, zero-shot relative error, and the discretization defect $\|D_R f_{2R}(U_R x) - f_R(x)\|$. This experiment asks whether crossing the theoretical threshold predicts a stable training intervention and resolution transfer, beyond matched base-grid accuracy.

4.4 LATENT ENCODERS EXPOSE AN IMPLICIT FILTERING REQUIREMENT

We do not assume that compression automatically supplies operator-ideal regularity. On an antialiased image-resolution ladder, we separately estimate the encoder pushforward $K_{E,N}$ in Equation (23) and the actual relative diffusion update $K_{\text{diff},N,t}$ in Equation (25). The controls are pixel identity, a same-rank partial isometry, architecture-matched random encoders, antialiased downsampling, a learned VAE encoder, and the learned encoder with an explicit Schatten gate. The principal ladder

allows latent spatial rank to grow with resolution; fixed-rank bottlenecks are reported separately and are not treated as evidence of nontrivial uniform $\mathcal{S}_2$ control.

The audit reports full covariance spectra, raw $T_2(N)$ and its growth exponent, effective rank, tail energy, and downstream noise-scale or gradient drift. The latent reference covariance is estimated independently of the injected perturbation, with a fixed support regularization rule across the ladder. Raw T_2 under the quadrature-weighted discretization inner product is the membership test; normalized effective-rank summaries are complementary descriptions, not substitutes. This experiment answers the representation question directly: which encoders change the spectral class in which noise acts, and when explicit filtering supplies what compression alone does not.

5 RELATED WORK

Noise injection and diffusion. Dropout established stochastic feature perturbation as a general-purpose regularizer (Srivastava et al., 2014), while diffusion models make repeated Gaussian perturbation the central learning mechanism (Ho et al., 2020). Latent diffusion moves this process into an autoencoded representation to reduce computation while retaining perceptually important information (Rombach et al., 2022). Our question is orthogonal to the nominal noise schedule: whether its *relative covariance family* remains uniformly controlled as representation resolution grows, and whether the encoder or an explicit gate supplies that control.

Function-space probabilistic learning. Function-space diffusion and variational neural operators formulate generative learning directly on infinite-dimensional spaces (Kerrigan et al., 2023; Seidman et al., 2023); recent work also develops well-defined regularized variational objectives for function-space Bayesian inference (Cinquin & Bamler, 2025). We study a complementary object: the resolution trajectory of finite noise-injection objectives, and a computable criterion that determines whether a nominal perturbation remains the same geometric intervention under refinement.

Neural operators and discretization. Fourier neural operators are designed to share parameters across discretizations (Li et al., 2021). That property does not by itself make a stochastic training intervention discretization independent: a normalized Fourier mask may add new noise mass whenever the grid exposes new modes. We use operator learning as the controlled test bed in which the underlying function, physical modes, and task remain fixed while only the cutoff changes.

Gaussian operator geometry. Gaussian equivalence theory and Hilbert–Carleman determinants identify Hilbert–Schmidt perturbations as the natural domain for comparing infinite-dimensional Gaussian measures (Bogachev, 1998; Minh, 2022). We use this machinery as an explanation and turn it into a learning-facing theorem, matrix-free diagnostic, and spectral gate. The protagonist is resolution-dependent training instability rather than measure theory itself.

Spectral regularization. Spectral normalization controls a largest singular value (Miyato et al., 2018); low-pass and low-rank methods constrain coordinate frequency or finite rank. None of these summaries alone determines cumulative square summability of a whitened relative covariance. Schatten-filtered noise targets that quantity uniformly across discretizations. The matched-frequency construction in Section 4.2 holds the Fourier diagonal, trace, and operator norm fixed while varying only the hidden multiplicity that determines T_2 .

6 CONCLUSION

Gaussian noise can be small in every resolved direction and still become a large perturbation as resolution grows. This paper identifies the missing criterion: dimension-uniform $\mathcal{S}_2$ control is the sharp threshold for refinement-stable Gaussian covariance objectives and their sample-level fluctuations. Schatten-filtered noise makes that requirement explicit, while the Carleman–Fredholm determinant supplies the correct objective in the Hilbert–Schmidt but non-trace-class regime. The resulting view explains why frequency attenuation and dimensionality reduction alone do not determine continuum stability, and turns the operator-ideal role of latent encoders into a measurable hypothesis. Noise injection should therefore be designed and audited as a resolution-indexed spectral object, not as a dimension-free scalar hyperparameter.

AI USE STATEMENT

Generative-AI tools assisted with outlining, prose drafting, code scaffolding, and consistency checks. The authors inspect the resulting text and code, independently verify mathematical and empirical claims, and retain full responsibility for the submitted work.

ETHICS STATEMENT

This work develops mathematical diagnostics and learning algorithms and does not involve human participants, private data, or deployment decisions. We do not identify material ethical risks beyond the ordinary resource and misuse considerations associated with machine-learning research.

REPRODUCIBILITY STATEMENT

The appendix specifies the proofs, estimators, matched controls, and discretization protocol. The accompanying artifact contains executable scripts, raw CSV outputs, figure generation, environment information, seeds, and provenance records for every reported result.

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A PROOF OF THE STABILITY THEOREM

Let $\lambda \in [0, M]$ and $g(\lambda) = \lambda - \log(1 + \lambda)$. Since

$$g(\lambda) = \int_0^\lambda \frac{t}{1+t} dt, \quad (33)$$

we have

$$\frac{\lambda^2}{2(1+M)} \leq g(\lambda) \leq \frac{\lambda^2}{2}. \quad (34)$$

Summing over the eigenvalues of K_N and multiplying by $1/2$ proves Equation (16).

In whitened coordinates, write $X = (I + K_N)^{1/2} \xi$ with $\xi \sim \mathcal{N}(0, I)$. Diagonalizing K_N gives

$$\ell_N(X) - \mathbb{E}_{q_N} \ell_N(X) = \frac{1}{2} \sum_j \lambda_{j,N} (\xi_j^2 - 1). \quad (35)$$

The terms are independent and $\text{Var}(\xi_j^2) = 2$, hence

$$\text{Var}_{q_N}[\ell_N(X)] = \frac{1}{2} \sum_j \lambda_{j,N}^2 = \frac{1}{2} T_2(N). \quad (36)$$

The signed case follows by bounding $g(\lambda)/\lambda^2$ on the compact interval $[-1 + \delta, M]$.

B MATRIX-FREE ESTIMATORS

Let $A_N = K_N^* K_N$. For an unnormalized Rademacher probe v , $\|K_N v\|^2 = v^\top A_N v$ and

$$\mathbb{E}[v^\top A_N v] = \text{Tr } A_N = T_2(N), \quad \text{Var}[v^\top A_N v] = 2 \sum_{i \neq j} (A_N)_{ij}^2. \quad (37)$$

The variance decreases by a factor of m after averaging m independent probes. For the series in Equation (22), if $\|K_N\|_{\text{op}} \leq \rho < 1$ and terms through order L are retained, then

$$\left| -\log \det_2(I + K_N) - \sum_{k=2}^L \frac{(-1)^k}{k} \text{Tr}(K_N^k) \right| \leq \frac{\rho^{L-1}}{(L+1)(1-\rho)} T_2(N). \quad (38)$$

The estimator requires $O(mL)$ operator–vector products and supports backpropagation through the same computation. All controlled-spectrum tests compare probe estimates against exact eigenvalue sums; the frequency-matched construction additionally validates the formulas with explicit DFT matrices.

C FREQUENCY-MATCHED CONTROLS

For completeness, we verify the identities used in Section 4.2. A flat unitary has constant-modulus entries, so each diagonal entry of $U \text{diag}(\lambda) U^*$ equals $d^{-1} \sum_j \lambda_j$. Both spectra in Equation (29) have trace ηm , giving the shared diagonal η/m . Their largest eigenvalue is η , while direct summation gives

$$\sum_j (\lambda_j^{\text{stable}})^2 = \eta^2 \left[1 + \frac{m^2 - 1}{(m+1)^2} \right] = \eta^2 \frac{2m}{m+1}, \quad (39)$$

$$\sum_j (\lambda_j^{\text{unstable}})^2 = m\eta^2. \quad (40)$$

Thus the stable family has bounded Hilbert–Schmidt mass but diverging trace, whereas the unstable family leaves $\mathcal{S}_2$.

To match spatial-frequency appearance, let

$$G_R = \text{diag} \left[(1 + \|k\|^2)^{-s/2} \right]_{\|k\| \leq R}, \quad s > 1, \quad (41)$$

and set $K_{R,d}^{(\cdot)} = G_R \otimes A_d^{(\cdot)}$. The two families have the same Fourier diagonal and hence the same radial power summary. For every spatial mask M_R ,

$$\left\| (M_R G_R M_R) \otimes A_d^{(\cdot)} \right\|_{S_2}^2 = \|M_R G_R M_R\|_{S_2}^2 \left\| A_d^{(\cdot)} \right\|_{S_2}^2. \quad (42)$$

Any frequency-only control that retains a nonzero low-frequency response has $\|M_R G_R M_R\|_{S_2} \geq c > 0$ and therefore preserves the unstable family’s $\sqrt{d}$ divergence. The comparison matches Fourier diagonal, radial power, trace, operator norm, bandwidth, low-frequency response, parameter count, probe count, and compute budget. Closed-form spectra provide the ground truth; Hutchinson probes only validate the matrix-free estimator.

D ADDITIONAL EXPERIMENTAL DETAILS

The final artifact records full resolution ladders, seeds and confidence intervals, hyperparameters, compute, estimator probe counts, discretization weights, encoder scaling constants, and provenance hashes. Operator-learning fields are generated once at the highest resolution and restricted to each grid, preventing task variation from being confounded with refinement.